J. WAP to add two 8-bit numbers stored in memory location.

Algorithm

① Load addend and augend into registers H and accumulator
② Add A = A + H
③ Store the result in memory (9100H)
④ If (carry)
    A = 1
    Store ~~accum~~ content in memory (9101H)
⑤ Stop

Program

```
MVI  B, 00H   ; remove garbage value
LXI  H, 9001H ; load ~~av addend~~ H-register with 9001H
LDA  9000H    ; load accumulator with data in 9000H
ADD  M        ; A = A + M
STA  9100H    ; store content of accumulator in 9100H
CC   carry    ; in case of carry
HLT
carry:
MVI  A, 01H   ; store 01H in accumulator
STA  9101H    ; store content of accumulator in 9101H
RET
```

Output

9000	FF
9001	02

9100	10
9101	01

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2. WAP to substract two 8-bit numbers stored in memory location.

Algorithm

① Load substrahend and minuend in H-register and Accumulator

② Substract A = A − H

③ If (carry)

A = FFH − A

A = A + 1

Store content in memory (9100H)

A = 01H

Store content in memory (9101H)

④ Stop

Program

```
LHLD  H, 9001H     ; minuend in memory ~~HL = 9001H~~ loaded in HL pair
LDA   9000H        ; load accumulator with substrahend in 9000H
SUB   H            ; A = A − H
STA   9100H        ; store content of accumulator with 9100H
CC    borrow       ; call if result is negative
HLT
borrow:
CMA                ; complement content of accumulator
ADI   01H          ; A = A + 1
STA   9100H        ; store content of accumulator in 9100H
MVI   A, 01H       ; A = 01H
STA   9101H        ; store content of accumulator in 9101H
RET
```

Output

9000	CD
9001	FF

9100	32
9101	01

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3. WAP to add two 16-bit ~~numb~~ numbers stored in memory location.

Algorithm

① Load 16-bit content into HL-register pair.
② ~~No~~ Exchange contents of DE and HL-register pair.
③ Load 16-bit ~~ad~~ augend into ~~DE regis~~ HL-register pair.
④ DAD HL = HL + DE
⑤ Store result in memory (9100H)
⑥ If (carry)
  A = ~~A~~ ~~00~~ 01H
  Store result in memory (9102H)
⑦ Stop

Program

~~MVI B, 00H ; remove garbage value~~
~~LXI H, 9001H ; load H-register with 9~~
LHLD 9000H ; load HL-reg. pair with contents of 9000H
XCHG ; exchange contents of HL and DE reg. pair
LHLD 9002H ; load HL-reg. pair with contents of 9002H
DAD D ; HL = HL + DE
SHLD 9001H ; store content of HL-reg. pair in 9001H
CC carry ; call if carry
HLT
carry :
MVI A, 01H ; A = 01H
STA 9102H ; store content of accumulator in 9102H
RET

Output

9000	CD
9001	FF
9002	33
9003	11

910~~0~~0	00
910~~2~~1	11
9102	01

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4. WAP to substract two 16-bit numbers stored in memory location.

Algorithm

1) ~~Load minuend and substraend into HL~~
1) Load ~~higher order~~ lower order byte of minuend in accumulator
2) Load 16-bit substrahend into HL-reg. pair
3) A = A - L
4) Load ~~lower or~~ higher order byte of minuend in accumulator
4) A = A - H
5) Store the result in memory (9101H)
6) Stop

Program

```
LHLD  9002H  ; load HL reg. pair with content of 9002H
LDA   9000H  ; load lower order byte of minuend in accumulator
SUB   L      ; A = A - L
STA   9100H  ; store content of accumulator in 9100H
LDA   9001H  ; load higher order byte of substrahend in accumulator
SBB   H      ; A = A - H - C
STA   9101H  ; store contents of accumulator in 9101H
HLT
```

Output

9000	CD
9001	FF
9002	33
9003	11

9100	9A
9101	EE

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5. WAP to multiply two 8-bit numbers stored in memory location.

Algorithm

~~① Load first number into B-register~~

~~② Load second number into~~

① Load numbers to multiply into HL reg. pair

② Add A = A + H

③ If (carry)

D = D + 1

④ Sub L = L - 1

⑤ If (L ~~=0~~ != 0)

Go to ②

⑥ Stop

Program

```
MVI D, 00H ; remove garbage value
MVI A, 00H ; remove garbage value
LHLD 9000H ; load both numbers to multiply in HL-reg. pair
multiply:
ADD H ; A = A + H
CC carry ; call if carry
DCR L ; L = L - 1
JNZ multiply ; multiplication not complete then jump
STA 9100H ; store content of accumulator in 9100H
MOV A, D ; move content of D-register in accumulator
STA 9101H ; store content of accumulator in 9101H
HLT
carry:
INR D ; D = D + 1
RET
```

Output

9000	EE
9001	02

9100	DC
9101	01

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6. WAP to divide two 8-bit numbers stored in memory location.

Algorithm

1. Load divisor into B-register
2. Load accumulator with dividend
3. C = 00H
4. ~~Com~~ If (A < B)
   Store content of accumulator in memory (9101H)
   Store content of C-register in memory (9100H)
5. Sub A = A - B
6. C = C + 1
7. Go to 4 if (A > B)
8. Stop

~~B~~

Program

```
LDA 9000H   ; load accumulator with ~~dividend~~ divisor
MOV B, A    ; move content of B-register into accumulator
LDA 9001H   ; load accumulator with dividend
MVI C, 00H  ; remove garbage value
loop:
CMP B       ; Compare A and B
JC end      ; if A < B, then r = divisor q = 0
SUB B       ; A = A - B
INR C       ; C = C + 1
JMP loop    ; unconditional loop
end:
STA 9101H   ; store remainder in memory 9101H
MOV A, C    ; move content of C-register to accumulator
STA 9100H   ; store quotient in memory 9100H
HLT
```

Output

9000	07
9001	CD

9100	1D
9101	02

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